Journal of Membrane Science 730 (2025) 124168

Contents lists available at ScienceDirect

Journal of Membrane Science

journal homepage: www.elsevier.com/locate/memsci

Fractal analysis of porous structure and permeability of separation membranes

Riccardo Lovison [a], Mohammad A. Afzal [b], Christina M. Carbrello [c], Andrew L. Zydney [b], Yifu Ding [a,*]

[a] Membrane Applications Science and Technology (MAST) Center, Paul M. Rady Department of Mechanical Engineering, University of Colorado Boulder, Boulder, CO, 80309-0427, USA
[b] Membrane Applications Science and Technology (MAST) Center, Department of Chemical Engineering, The Pennsylvania State University, University Park, PA, 16802, USA
[c] MilliporeSigma, 80 Ashby Rd. Bedford, MA, 01730, USA

ARTICLE INFO

Keywords:
Fractal dimensions
Porous membranes
Pore size distribution
Permeability

ABSTRACT

Pore structures of membranes are routinely characterized by imaging techniques. However, it remains a challenge to quantify the images, and ultimately, to correlate the calculated geometric quantities with membrane performance. In this study, we present systematic fractal analysis of images of membrane pore structures. Both pore-area (D_f) and tortuosity fractal dimensions (D_T) were obtained from in-plane and in-thickness cross-sectional SEM images, respectively. Analysis of symmetric PVDF membranes reveal that the best method of extracting fractal dimension value is the box counting method (BCM) with box size range between minimum (λ_{min}) and maximum (λ_{max}) pore size. Importantly, the permeability of the membranes can be accurately calculated using geometric parameters (λ_{max}, porosity (ϕ), D_f, and D_T) that are all obtained from the images. We further applied the methodology to two asymmetric PES membranes by approximating the membranes as having separate layers each with different pore structures. The geometric parameters including fractal dimensions were determined for these layers, which allowed estimations of the permeability of these layers. Using resistance-in-series approach, the calculated overall membrane permeance matches well with the experimental results, with the first 2–3 μ m layer contributing over 66 % resistance for both membranes.

1. Introduction

Nomenclature

List of symbols

A	Surface area
A_P	Pore Area
a	Island size (variable)
a_{max}	Maximum island size
C_k	Kozeny-Carman Constant
C_i	Contribute to the resistance
d	Euclidean dimension
D	Fractal dimension
D_f	Pore-area fractal dimension
D_T	Tortuosity Fractal dimension
$f(\lambda)$	Pore size distribution
$g(\lambda)$	Probability density function
K	Permeability (Darcy)
K_{exp}	Experimental permeability
K_{FPM}	Permeability predicted via FPM
l_0	Thickness of the membrane
l_t	Tortuous path length
Δm	Mass variation
$M(L)$	Length (or Area, or volume) of fractal objects
$N(A \geq a)$	Number of islands with size A larger than a
$N(L \geq \lambda)$	Number of pores larger than λ
N_t	Total number of pores
Δp	Pressure drop
Pe	Permeance (Darcy)
$Q(\lambda)$	Flux through a capillary
Q_{exp}	Experimentally measured flux
Q_{total}	Total flux across the membrane
R_i	Resistance of the layer i

(continued on next column)

(continued)

(continued on next page)

* Corresponding author.
E-mail address: Yifu.ding@colorado.edu (Y. Ding).

https://doi.org/10.1016/j.memsci.2025.124168
Received 6 February 2025; Received in revised form 18 April 2025; Accepted 30 April 2025
Available online 5 May 2025
0376-7388/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(*continued*)

R_{tot}	Total resistance of the membrane
S_v	Specific surface area
t	Time
Greek letters	
ϵ	Length Scale
λ	Pore size
λ_{max}	Maximum pore size
λ_{min}	Minimum pore size
μ	Viscosity
ρ	Density
τ	Tortuosity through single channel
$\tau_{average}$	Average tortuosity
ϕ	Porosity
Subscripts	
PES_200	Polyether sulfone 200 nm pore size rating
PES_20	Polyether sulfone 20 nm pore size rating
PVDF_650	Polyvinylidene fluoride 650
*PVDF*_100	Polyvinylidene fluoride 100
s_{BCM}	Standard Box Counting Method
t_{BCM}	Modified Box Counting Method
D_{s_BCM}	Fractal Dimension obtained with s_BCM
D_{t_BCM}	Fractal Dimension obtained with t_BCM
D_{PSD}	Fractal Dimension obtained from PSD
D_{Th}	Fractal Dimension obtained from Eq. 25
IPCS	In-plane cross-section
ITCS	In-thickness cross-section

Pore structures dictate the most relevant properties of membranes including permeability and selectivity. The ability to characterize pore structures is therefore crucial for both understanding membrane properties and developing high-performance membranes to meet specific needs from wastewater treatment to virus removal. In biomanufacturing, this is especially important given the broad range of particle size in the feeds [1]. For example, Taylor et al. [2] correlated particle transmission with the ratio of the particle diameter to either the maximum pore size or the mean pore size. For applications like membrane distillation, the internal pore structure affects the wetting of the membrane, vapor flux, and even heat transfer across the membrane [3–5].

Pore structures are routinely characterized by real space imaging techniques that typically yield 2D images of the membrane. Three dimensional images with different resolutions can be obtained using techniques like FIB-SEM [6] or confocal microscopy [7]. For example, 3D structural information with sub-nanometer resolution was obtained for the polyamide barrier layer on Reverse Osmosis (RO) membranes using FIB-SEM [8]. However, it is often challenging to quantify these images and correlate the obtained geometric parameters with membrane performance.

On the other hand, membrane pore structures can be evaluated with pressure-driven intrusion of gas or liquid, from techniques such as bubble point method [9], gas-liquid displacement porometry [10] (GLDP), liquid-liquid displacement porometry [11] (LLDP), and mercury intrusion porosimetry [12]. Generally, GLDP faces challenges in determining small pore sizes [13], while LLDP experiences challenges in accurately measuring maximum pore sizes and correlating it with virus retention [11,14–16]. Quantification of pore size distribution (PSD) can be further improved by measuring the retention of nanoparticles [15] or solutes [17]. From these characterizations, membrane pore structures, especially those of microfiltration (MF) and ultrafiltration (UF) membranes, can be reasonably well described by the log-normal distribution [18]. Typically, pore size information obtained from these measurements is not compared with that from the images. A further limitation of these techniques is that they are only available to provide PSD of the tightest upper part of membranes, without giving insights into the underneath porous substrate.

Nevertheless, quantifications of the pore structure, including PSD, porosity and pore interconnectivity (tortuosity), obtained from abovementioned characterization methods, can be applied to predict permselective properties of the membrane. Starting with the classical Hagen-Poiseuille (HP) relationship, the volumetric flow rate of a single capillary with diameter λ, can be estimated,

$$Q(\lambda) = \frac{\pi \Delta p \lambda^4}{128 \mu l_0} \tag{1}$$

where μ is the fluid viscosity, Δp is the applied pressure, and l_0 is the capillary length (neglecting the effects of tortuosity). Applying the log-normal PSD, $f(\lambda)$, to the different sized capillaries, the total volumetric flow rate can then be obtained,

$$Q_{total} = \int_{\lambda_{min}}^{\lambda_{max}} Q(\lambda) \cdot f(\lambda) d\lambda \tag{2}$$

where λ_{min} and λ_{max} represent the diameters of the smallest and the largest capillaries. Finally, the intrinsic permeability (K) of the membrane can be obtained from Darcy's law.

$$K = \frac{Q_{total} \mu l_0}{\Delta p A} \tag{3}$$

where A represents the active area of the membrane. Building on such analysis, Mehta and Zydney [19] obtained a correlation between permeability and selectivity for commercial UF membranes. Further, Siddiqui et al. [20] investigated the impacts of pore shape, represented by the pore aspect ratio, and PSD on the permselective properties of the membranes.

Another approach, the Kozeny-Carman (KC) equation, is commonly used to connect the K of a porous medium with the pore structure characteristics,

$$K = \frac{\phi^3}{C_k \tau^2 (1-\phi)^2 S_v^2} \tag{4}$$

where ϕ, τ, S_v represent porosity, tortuosity, and specific surface area of the porous medium, and C_k is the KC constant. The values of τ, S_v and C_k are typically used as fitting parameters for analysis of the membranes [21].

Fractal analysis, a geometric approach, has been used to characterize the self-similar structure of random porous media such as soils [22]. The permeability of a porous medium can be calculated using a set of fractal dimensions based on the pore size and porosity [23]. Xu et al. derived an analytical expression for the KC constant in Eq. (4) using fractal analysis [24], which provides improved description of the permeability of both mono-dispersed and bi-dispersed porous media. Cai et al. [25] introduced a model to estimate the maximum pore size of the porous medium based on fractal geometry, finding a good agreement between the predicted values and the experimental data. Furthermore, Cai et al. [26] successfully applied the fractal analysis to describe the imbibition kinetics of a wetting liquid into a gas-saturated porous medium.

A few studies have explored the use of fractal analysis to characterize membranes. He et al. [27] applied fractal geometry to predict the permeability of the gas diffusion layer in a proton exchange membrane for fuel cell applications. Zhang [28] developed a similar fractal-based permeation model to describe gas transport through porous PVDF membranes. Recently, Win et al. [29] used the fractal dimension to quantify the simulated morphology of a microphase separated terpolymer as a proton exchange membrane. The diffusion of water and conductivity through the membrane increases with the increase of fractal dimensions of the microstructure. Martinez et al. showed that the fractal-dimension based model provides better description of the capillary imbibition of porous membranes [30].

Despite the abovementioned efforts, it remains unclear as to how to reliably quantify the fractal dimensions of porous membranes and subsequently, how effective the fractal dimensions are in describing the permeability of common liquid-separation membranes. Here, we

systematically investigated the quantification and application of fractal dimensions for common MF membranes, purely based on images of the membranes. For a given membrane, the fractal dimension values, including both pore-area and tortuosity fractal dimensions, were obtained using four different approaches. Subsequently, the as-obtained fractal dimensions were incorporated into a fractal permeability model (FPM) to predict the permeability/permeance of the membrane. Both symmetric polyvinylidene fluoride (PVDF) membranes and asymmetric polyether sulfone (PES) membranes, each with two different pore sizes, were analyzed.

2. Fractal permeability model

As defined by Mandelbrot, "A fractal is a rough or fragmented geometric shape that can be split into parts, each of which is (at least approximately) a reduced-size copy of the whole" [31]. A fractal, or a self-similar object, displays a characteristic scaling relationship [32],

$$M(L) \sim \epsilon^D \tag{5}$$

where M can be the length, the area or the volume of an object, ϵ is the length scale, and D is the fractal dimension of the object. For classical fractals such as the Sierpinski gasket, the Koch curve, and the Apollonian gasket, the value D is constant over an infinite range of length scale [33].

Real objects such as random porous media can display statistical self-similarity within a limited range of length scale [34]. This is inspired by Mandelbrot's description of the distribution of islands on earth [31],

$$N(A > a) \propto a^{\frac{D}{2}} \tag{6}$$

where N is the number of islands with size A larger than a. Majumdar et al. [35] introduced the same concept for describing the contact spots on engineered surfaces,

$$N(A \geq a) \propto \left(\frac{a_{max}}{a}\right)^{\frac{D}{2}} \tag{7}$$

where a_{max} is the maximum area of contact spots. Yu and Li [34] introduced a unified model for fractal character of random porous media,

$$N(L \geq \lambda) = \left(\frac{\lambda_{max}}{\lambda}\right)^D \tag{8}$$

where N is the number of pores larger than the pore size λ and λ_{max} is the maximum pore size. Thus, the total number of pores in between λ_{max} and the minimum pore size (λ_{min}), N_t, is,

$$N_t(L \geq \lambda_{min}) = \left(\frac{\lambda_{max}}{\lambda_{min}}\right)^D$$

It is worth noting that this analysis assumes the PSD of the porous media follows a "fractal" power-law distribution, rather than the commonly used log-normal distribution for membranes from porometry and/or rejection tests [18]. Subsequently, the porosity of the media (ϕ) correlates with the pore sizes and fractal dimension,

$$\phi = \left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{d-D} \tag{10}$$

where d is the Euclidean dimension: $1 < D < 2$, if $d = 2$; or $2 < D < 3$, if $d = 3$. By differentiating Eq (8), the probability density function ($g(\lambda)$) describing the pore size distribution can be obtained for pore-area ($d = 2$) fractal dimension D_f,

$$-\frac{dN}{N_t} = D_f \lambda_{min}^{D_f} \lambda^{-D_f-1} d\lambda = g(\lambda) d\lambda \tag{11}$$

with $g(\lambda) \geq 0$ and $\int_0^\infty g(\lambda)d\lambda = \int_{\lambda_{min}}^{\lambda_{max}} g(\lambda)d\lambda = 1$, which can be only satisfied if,

$$\left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{D_f} \cong 0 \tag{12}$$

In the same paper, the authors consider this assumption as correct when $\lambda_{min}/\lambda_{max}$ is smaller than 10^{-2}.

The second fractal dimension used to describe the porous media is the so-called tortuosity fractal dimension (D_T), which describes the complexity of the interconnectivity of pores throughout the thickness. Approximating the porous media as tortuous capillary bundles with varying diameter (λ), D_T is then adopted to define the capillary length $l_t(\lambda)$ [36],

$$l_t(\lambda) = \lambda^{1-D_T} l_0^{D_T} \tag{13}$$

where l_0 is the length of the straight capillary that equals the thickness of the porous medium. Correspondingly, the tortuosity factor, as commonly used in the membrane field, can be correlated with D_T,

$$\tau(\lambda) = \frac{l_t(\lambda)}{l_0} = \left(\frac{l_0}{\lambda}\right)^{D_T-1} \tag{14}$$

Combining Eq. (11) and (12) and Eq. (14) and integrating through minimum and maximum pore size, it is possible to estimate the average tortuosity, $\tau_{average}$, as

$$\tau_{average} = \int_{\lambda_{min}}^{\lambda_{max}} \tau(\lambda) f(\lambda) d\lambda = \frac{D_f}{D_f + D_T - 1} \left(\frac{l_0}{\lambda_{min}}\right)^{D_T-1} \tag{15}$$

Based on above fractal dimensions, permeability (K) of random porous media can be modelled [28,37,38], as briefly discussed below.

The volumetric flow can be measured as in Eq. (1), substituting l_0 with l_t from Eq. (13). Then, the overall flux can be obtained integrating over the fractal pore size distribution [39], Eq. (11), as

$$Q_{total} = -\int_{\lambda_{min}}^{\lambda_{max}} Q(\lambda) dN(\lambda) \tag{16}$$

$$Q_{total} = \frac{\pi}{128} \frac{\Delta p}{\mu} \frac{A}{l_0} \frac{l_0^{1-D_T}}{A} \frac{D_f}{3 + D_T - D_f} \times \lambda_{max}^{3+D_T} \left[1 - \left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{D_f} \left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{3+D_T-2D_f}\right] \tag{17}$$

With $1 < D_f < 2$ and $1 < D_T < 2$, the exponent $3 + D_T - 2D_F > 0$. In the limit of Eq. (12),

$$Q_{total} = \frac{\pi}{128} \frac{\Delta p}{\mu} \frac{A}{l_0} \frac{l_0^{1-D_T}}{A} \frac{D_f}{3 + D_T - D_f} \lambda_{max}^{3+D_T} \tag{18}$$

Note that the absence of any dependency of Q_{total} on λ_{min} in Eq. (18), and in the subsequent analysis, is due to the assumption that λ_{min} is very small based on Eq. (12). Substituting Eq. (18) in Eq. (3),

$$K = \frac{\mu l_0 Q_{total}}{\Delta p A} = \frac{\pi}{128} \frac{l_0^{1-D_T}}{A} \frac{D_f}{3 + D_T - D_f} \lambda_{max}^{3+D_T} \tag{19}$$

For a fractal-like porous medium, the actual pore area is [24],

$$A_p = -\int_{\lambda_{min}}^{\lambda_{max}} \frac{\pi}{4} \lambda^2 dN = \frac{\pi D_f}{4(2 - D_f)} \lambda_{max}^2 [1 - \phi] \tag{20}$$

Accordingly, the effective area of the porous medium is,

$$A = \frac{A_p}{\phi} = \frac{\pi D_f}{4\phi(2 - D_f)}\lambda_{max}^2[1 - \phi] \tag{21}$$

Next, assuming a symmetric pore structure, l_0 is estimated as,

$$l_0 = \sqrt{\frac{A_p}{\phi}} = \left[\frac{\pi}{4}\frac{D_f}{2 - D_f}\frac{1 - \phi}{\phi}\right]^{\frac{1}{2}}\lambda_{max} \tag{22}$$

Finally, combining Eqs. [19 , 22], the theoretical permeability of a porous medium can be calculated based on purely structural and geometrical parameters of the medium, ϕ, D_f and D_T, and λ_{max} as follows:

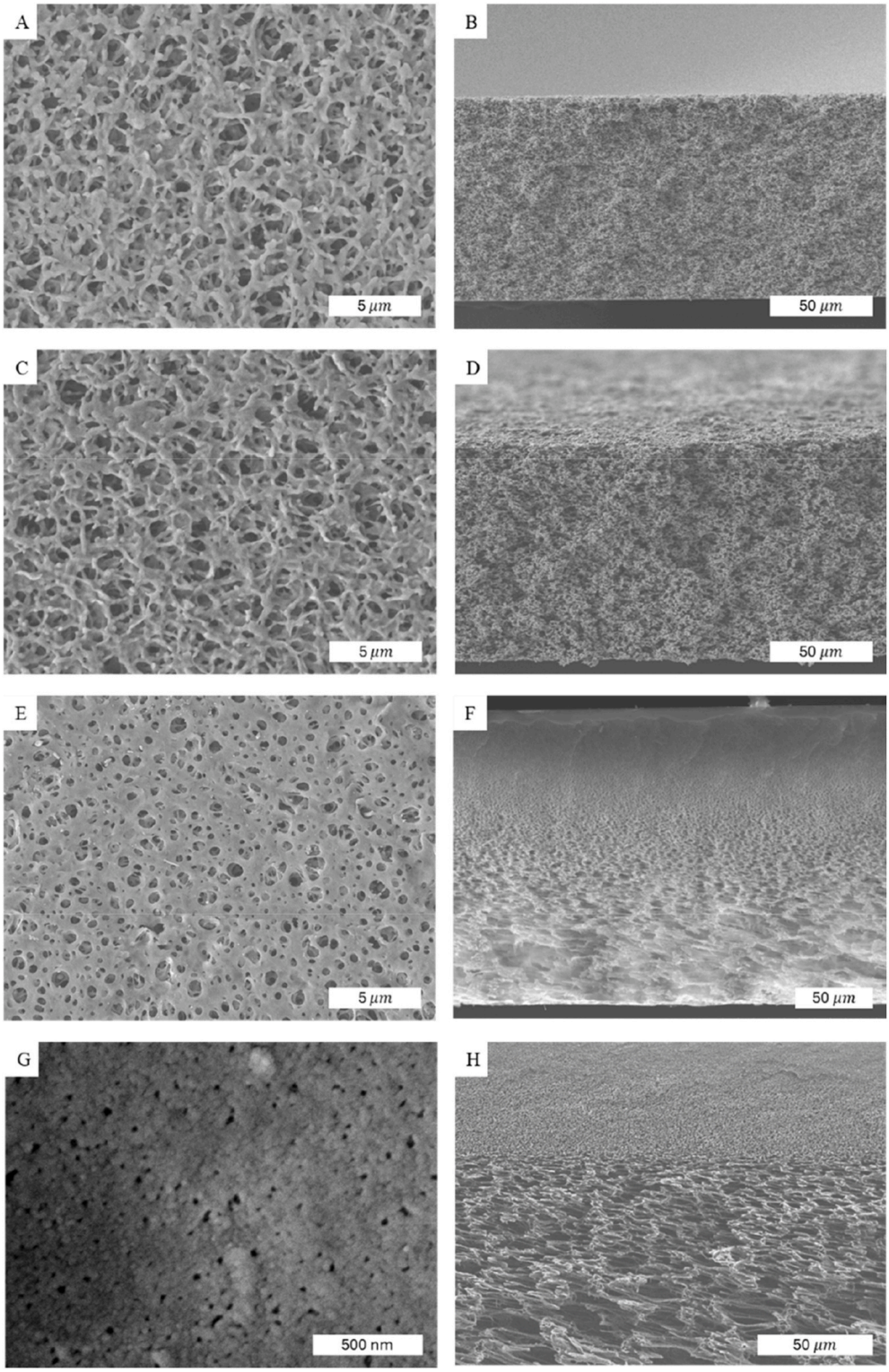


Fig. 1. SEM images of the membranes. (A) PVDF_100 surface, (B) PVDF_100 cross-section, (C) PVDF_650 surface, (D) PVDF_650 cross-section, (E) PES_200 surface, (F) PES_200 cross-section, (G) PES_20 surface, (H) PES_20 cross-section.

$$K_{FPM} = \frac{\pi}{128}\left(\frac{D_f}{3 + D_T - D_f}\right)\left[\frac{\pi}{4}\frac{D_f}{2 - D_f}\frac{1-\phi}{\phi}\right]^{-(D_T+1)/2}\lambda_{max}^2 \quad (23)$$

In this study, fractal dimensions estimated from four different approaches are compared using symmetric PVDF membranes. The applications of the fractal dimensions in predicting the permeance of both PVDF membranes and asymmetric PES membranes are demonstrated.

3. Materials and methods

3.1. Membranes

Both the PVDF and the PES membranes were obtained from MilliporeSigma. PVDF membranes have symmetric pore structures across the thickness [40,41] (Fig. 1A–D). In contrast, PES membranes display characteristic asymmetric pore structures, resulting from the nonsolvent induced phase separation (NIPS) fabrication process [30,42], as shown in Fig. 1E–H.

Table 1 lists the relevant properties of all four membranes including, the overall porosity ($\phi_{overall}$), nominal pore size rating ($\bar{\lambda}$), membrane thickness (l_0), and their DI water permeance (Pe_{exp}). $\phi_{overall}$ was measured using gravimetric measurements [43], $\bar{\lambda}$ was provided by the manufacturer, and Pe_{exp} was measured using a dead-end filtration cell. All the membranes were received as flat sheets, stored in an ambient dry environment and subsequently used without further treatment unless otherwise specified.

3.2. Characterization of the membranes

3.2.1. Imaging pore structures

The images of the PVDF membranes used for fractal analysis were obtained with scanning electron microscopy (SEM) and field-emission scanning electron microscopy (FE-SEM, JEOL JSM-7401F, Japan). The cross-sections of the samples were prepared by fracturing in liquid nitrogen. To minimize the charging effect, a 5 nm layer of platinum was deposited for all the samples with a sputter coater (Cressington 108 Auto).

For asymmetric PES membranes, the pore structure varies with depth. Recently, we developed a laser-ablation method to prepare in-plane cross-sections (IPCS) of the PES_200 membrane at different depths, as detailed in a companion article [45]. Briefly, controlled ablation of the PES membranes was achieved with a custom Femtosecond UV laser (Light Conversion CARBIDE-CB5, wavelength = 343 nm), coupled with an ALIO 6-D Hybrid Hexapod stage. To preserve the pore structure, a polypropylene (PP) with a molecular weight of 114 kg/mol was used as pore-preserving agent. By tuning the laser fluence and pulse number, different ablation depths were achieved for PES_200 membranes. A confocal microscope (Keyence VK-X 3D Surface Profiler) was then used to determine the depth of laser ablated regions. Finally, SEM images of the ablated regions with varying depths of the PES_200 membrane were obtained.

Images at different depths of PES_20 membrane were obtained via Focus-Ion-Beam SEM (FIB-SEM) following the general procedures described by Brickey et al. [44]. Samples were positioned with the size-selective exit side of the membrane facing outward. Before imaging, the membranes were sputter-coated with a 5 nm-thick iridium layer to prevent charging effects. Imaging was conducted using a dual-beam FEI Helios 660 FIB-SEM (ThermoFisher Scientific, Waltham, MA). During FIB-SEM tomography, the sample was tilted at an angle of 52°, aligning the membrane surface perpendicular to the milling direction. To protect the membrane surface from ion beam damage, a 100 nm-thick carbon layer was deposited using an electron beam, followed by a 1 μm-thick layer deposited with an ion beam for additional protection. Trenches were milled with the ion beam to expose a clean cross-section and reduce shadowing artifacts that could cause intensity variations in the images. Fiducial markers were introduced as reference points to ensure precise image alignment before and after milling for both ion and electron beam sites. Membrane cross-sections were milled using an ion beam set at 30 kV and 24 pA. SEM images were acquired at 5 kV and 0.2 nA using a through-lens detector (TLD). A total of 200 slices, each 7 nm thick, were collected for 3D reconstruction. After image acquisition, tilt and grayscale adjustments were made using Avizo (ThermoFisher Scientific). Vertical drifts were corrected with ImageJ [46]. Carbon deposition at the top of each image served as a template for vertical alignment, achieved with a custom-written software plugin [47]. Horizontal alignment was performed manually in Avizo, resulting in a properly reconstructed 3D volume. Surface images at different depths were extracted by adjusting the slice number along the vertical axis to visualize various planes within the 3D reconstruction.

Image analysis can be affected by subjectivity. For instance, using heavy image processing could generate artifacts and alterate the measured parameters, particularly small pores. For this reason, all the SEM images were processed using standard methods found in literature. First, a Gaussian filter was used to reduce noise. Then, an adaptive thresholding method [48] was used to obtain the first binarized image. An additional MATLAB code has been introduced to remove further noise from images. The final binarized images were used for calculating the fractal dimensions. Fig. S1 shows an example of image analysis at each of the processing steps for PVDF_100.

3.2.2. Permeation measurements

A stirred dead-end filtration cell (HP4750, Sterlitech, WA) with an active surface area of 11.5 cm^2 was used to determine the permeability/permeance of the membranes using DI water under N_2 pressurization. All experiments were conducted at room temperature. Before each permeation experiment, the membrane was soaked in an isopropanol/DI water solution (50:50 mixing ratio) for 2 min to pre-wet the membrane [41]. Next, the membranes were soaked in DI water for 2 min. The permeate mass flow rate ($\Delta m/t$) was measured with an automated electronic balance (PI-225DA, Denver Instrument) at 1-min intervals. The transmembrane pressure (Δp) used was 5 psi for PES_200 and PVDF_650, and 20 psi for PVDF_100. For each membrane, the permeability (K_{exp}) is experimentally determined from the steady-state water flow rate (Q), defined as the permeate flow rate with less than 2 % variation over a 5-min interval, using Darcy's law, as

$$K_{exp} = \frac{Q_{exp}\mu l_o}{\Delta p A} = \frac{\Delta m \mu l_0}{\rho t A \Delta p} = Pe\ \mu\ l_0 \quad (24)$$

where μ (=0.01 P) and ρ (=1 g/cm^3) are the viscosity and density of the DI water, l_0 and A are the thickness and active surface area of the membrane, respectively. Three samples of each membrane were measured, and the average permeance values are reported in Table 1.

Table 1
Properties of PES and PVDF membranes.

Membrane	$\phi_{overall}$ [%]	$\bar{\lambda}$ [nm]	l_0 [μ m]	Pe_{exp} [LMH/psi]
PES_20	73	20	140	40[a]
PES_200	80	200	175	920 ± 72
PVDF_100	60	100	90	177 ± 10
PVDF_650	64	650	110	1625 ± 50

[a] Value obtained from literature [44].

3.3. Determining fractal dimensions of membranes

The two parameters, pore-area fractal dimension (D_f, capturing the fractal-like feature in-plane) and tortuosity fractal dimension (D_T, capturing the fractal-like feature in the thickness direction), are calculated as follows. D_f was determined from SEM images of the surfaces of the PVDF membranes as they are symmetric. For PES_200 and PES_20, D_f values were obtained from the SEM images at different depths, as

prepared by laser ablation or FIB-SEM, respectively (Section 3.2.1.). Regarding D_T, SEM images of the cross section of the membranes were used. For symmetric structures, the D_T value has been considered constant across the cross-section (Fig. 1), whereas for asymmetric membranes different values corresponding to the top layer, characterized by tighter pores, and to the more porous sublayer have been estimated. Eq. (12) must be verified for each layer in order to apply FPM.

As porous membranes are statistically self-similar structures, the fractal dimension value will depend on the length-scale and methods of determination. Below, we compared four methods to estimate the values of D_f and D_T.

3.3.1. Methods to determine D_f

Method 1: Theoretical estimate ($D_{f,Th}$). Rearranging Eq. (10), the theoretical pore-area fractal dimension ($D_{f,Th}$) can be obtained as follows:

$$D_{f,Th} = 2 - \frac{\ln(\phi)}{\ln\left(\frac{\lambda_{min}}{\lambda_{max}}\right)} \tag{25}$$

In this case, the values of ϕ, λ_{min} and λ_{max} are obtained from the images of each membrane.

Method 2: PSD approach ($D_{f,PSD}$). For a given SEM image, the PSD is obtained, which is then fitted with a power-law function with varying fractal dimensions. The fractal dimension, $D_{f,PSD}$, is set equal to the value that provides the best fit (i.e., with the largest R^2 value).

Method 3: Standard Box Counting Method (s_BCM, D_{f,s_BCM}). This technique allows us to determine the fractal dimensions of the membranes directly from an image. BCM is commonly used in literature due to both its simplicity [49] and applicability for structures with or without self-similarity [49–53]. The analysis applies a grid of varying box sizes (with length, ϵ) to a given image and counts the number of boxes (N) that contain part of the characteristic feature (e.g. membrane pores). For a fractal-like object, $N \propto \epsilon^D$, and the fractal dimension can be determined as the slope of the $\log(N)$ vs $\log(\epsilon)$ plot,

$$D = \frac{\log(N)}{\log(\epsilon)} \tag{26}$$

The pore-area fractal dimension obtained with the standard BCM (using the entire box size range) is D_{f,s_BCM}.

Method 4: Truncated BCM (t_BCM, D_{f,t_BCM}). This method uses BCM with a limited box size range, between the minimum and maximum pore size. For well-defined fractals, the linearity of $\log(N) \sim \log(\epsilon)$ from standard BCM is found across any range of box sizes [54]. Practically, the self-similar feature starts to break down as the box size approaches the pixel size (lower limit) and frame size (upper limit) of the image, even for classical fractal images. For example, Fig. 2A shows an image of the Apollonian Gasket, a classical fractal object with a theoretical $D = 1.31$ [55]. Using the s_BCM on Fig. 2A, a $\log(N) \sim \log(\epsilon)$ plot can be obtained as shown in Fig. 2B, which yields $D = 1.40$ over the entire range of box sizes. However, Fig. 2C shows that the instantaneous slope of $\log(N) \sim \log(\epsilon)$ curve varies with the box size. Using the range of box sizes highlighted in Fig. 2C, a $D = 1.34$ was obtained, which agrees better with the theoretical value. At very small box sizes, the BCM struggles to capture the true fractal dimension due to the dependence on image resolution (contrast, quality, and pixel size) [56]. Conversely, as box size approaches the entire image frame, the BCM begins to capture features within each box, resulting in a dimension approaching 2, typical of Euclidean surfaces [57].

Such a box-size dependent fractal dimension could be even more evident for porous membranes, which only display self-similarity within a limited range of length scales. Accordingly, a sensitivity analysis of the BCM was carried out, as summarized in Fig. S2 and S3. Briefly, 2D images of "pores" with varying porosity, pore size and position distribution were generated and estimated with BCM. These analyses show that: (i) pore size impacts the range of box size for extracting the D values, but not the D values themselves (Fig. S2); (ii) porosity (ϕ) [58] and pore size range ($\lambda_{min}/\lambda_{max}$) impact the D values, consistent with Eq. (25) (Fig. S3A). In particular, the larger the porosity, the larger the measured D value. Furthermore, the broader the pore size distribution, the broader the range of box sizes where the structure shows fractal behavior (Fig. S3B). Based on all these analyses, t_BCM is explored in this study to quantify the D values for the membranes within a limited range of box size where the image is considered statistically self-similar.

3.3.2. Methods to determine D_T

Method 1: Eq. (15) is used to calculate $D_{T,Th}$, using τ values obtained from literature [59,60] for similar membranes. We emphasize that these τ values were not determined by the SEM images. Nevertheless, rearranging Eq. (15),

$$D_{T,Th} = \frac{D_f\left(\frac{l_0}{\lambda_{min}}\right)^{D_T-1} - \tau_{average}D_f + \tau_{average}}{\tau_{average}} \tag{27}$$

This implicit equation has been solved using numerical methods with MATLAB.

Methods 2–4: Analogous to the Methods 2–4 for determining D_f values (using in-plane cross-sectional SEM images) discussed above, $D_{T,PSD}$, D_{T,s_BCM} and D_{T,t_BCM} were obtained using in-thickness cross-sectional SEM images of the membranes. The use of t_BCM to determine D_{T,t_BCM} was validated using simplified images of tortuous capillaries (Fig. S4). The real tortuosity of capillaries was measured from the actual length of the channel divided by the linear-path length (Eq. (14)).

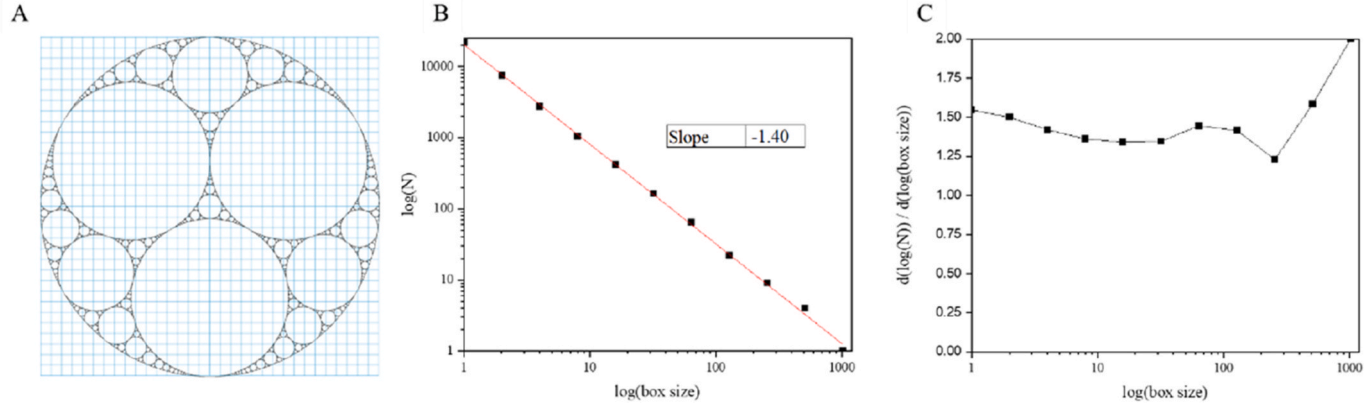


Fig. 2. (A) An Apollonian Gasket. (B) Number of boxes (N) containing a feature versus box size (ϵ) in a $log(N) \sim log(\epsilon)$ plot. (C) Derivative of (B) showing the variation of instantaneous fractal dimension with box size.

Alternatively, fractal tortuosity ($\tau_{fractal}$) was calculated using D_{T,t_BCM}. As shown in Fig. S4A, the two values agree well with each other.

4. Results and discussions

4.1. Fractal analysis of symmetric PVDF membranes

4.1.1. Pore-area fractal dimension

Two symmetric PVDF membranes (Fig. 1A–D) have been studied under the assumption that the main structural parameters, D_f, D_T, ϕ, λ_{min} and λ_{max}, remain constant throughout the membrane thickness. Table 2 summarizes the calculated values of D_f for both membranes, determined using the abovementioned four methods. Using PVDF_100 as an example, detailed analysis is described below. The analysis for PVDF_650 can be found in Supplementary Information (Fig. S5).

From the SEM image of PVDF_100 (Fig. 1A–B and Fig. S1), the PSD is obtained as shown in Fig. 3A. Accordingly, λ_{min}, λ_{max}, and ϕ are determined (Table 2), which were then used to calculate $D_{f,Th}$. Next, $D_{f,PSD}$ was obtained by fitting the PSD using a power-law function with varying exponents between 1.0 and 2.0 (at 0.1 intervals). From the best fit (the largest R^2 value in Fig. 3B), the value of $D_{f,PSD}$ was determined.

Fig. 4A shows the $\log(N)$ vs $\log(\epsilon)$ plot for the PVDF_100, comparing the results for the s_BCM and t_BCM methods. As explained in Section 3.3, the s_BCM applies a linear fit over the entire box size range to obtain D_{f,s_BCM}. In contrast, the t_BCM selects a specific range where the pore structure can be considered as statistically self-similar. As shown in Fig. 4B, the instantaneous D_f, i.e. the derivative of the data in Fig. 4A, clearly shows box-size dependency. The region where D_f is relatively constant (marked on Fig. 4B, defined as the region where the standard deviation between successive data points is below 0.15) was used to perform the linear fit and determine D_{f,t_BCM}. Note that the range identified using this method is consistent with the λ_{min} and λ_{max} values observed from the PSD (marked on Fig. 3A). Previously, Yu et al. [34] showed that the self-similarity of random porous media is valid within the range of λ_{min} and λ_{max}. The analysis here confirms that membranes also follow the same "range limitation". Applying BCM across entire range of box sizes, as used in other work [28,61–64], may lead to inaccurate quantification of D_f. Another advantage of using t_BCM is the reduced impact of image processing on the analysis. It is known that image processing can cause artifacts, which more significantly impact the value of λ_{min}. Here, the t_BCM defines $\lambda_{min} - \lambda_{max}$ as the range where D_f remains constant. Increasing λ_{min} to λ'_{min} will not affect the value of D_f (Fig. 4B).

For both membranes, $D_{f,PSD}$ is the lowest among the four pore-area fractal dimension values obtained. The values of $D_{f,Th}$ match reasonably well with D_{f,t_BCM}, as both methods emphasize the "range" of self-similar features of the pore structure. Compared with $D_{f,PSD}$, D_{f,t_BCM} captures both the PSD and the complexity of the pore connectivity (or location) in 2D. Li et al. [65] demonstrated that increasing the structural complexity will increase the fractal dimension value.

Between the two membranes, PVDF_650 showed slightly larger values of pore area fractal dimensions than PVDF_100 for three of the four methods (except D_{f,s_BCM}, which is significantly higher for PVDF_650). Given the similar porosity and pore size ratio of the two membranes, the similar D_f values are consistent with our sensitivity analysis of generated pore structures (Fig. S2–S3). Wu et al. [66] proposed that the difference between the fractal dimensions of two membranes can be attributed to the different pore complexity and their formation mechanisms. This could be true for the two PVDF membranes compared here. Lastly, the relatively large difference in D_{f,s_BCM} between the two membranes suggests that s_BCM more significantly overestimates the fractal dimension for PVDF_650. In comparison, Zhang [28] reported a $D_f = 1.94$ (and $D_T = 1.92$) for similar PVDF membranes using s_BCM. The analysis presented here suggests that t_BCM should be used for extracting the fractal dimension rather than the s_BCM.

Table 2
Pore-area fractal dimension obtained for PVDF_100 and PVDF_650.

	$\phi_{surface}$ [%]	λ_{min} [nm]	λ_{max} [nm]	$D_{f,Th}$	$D_{f,PSD}$	D_{f,s_BCM}	D_{f,t_BCM}
PVDF_100	40	50	550	1.62	1.36	1.56	1.51
PVDF_650	43	80	850	1.64	1.43	1.80	1.62

4.1.2. Tortuosity fractal dimension (D_T)

The four methods have also been applied to estimate the tortuosity fractal dimension of the two PVDF membranes. The results are listed in Table 3. As mentioned, the values of τ are needed to calculate $D_{T,Th}$ (Eq. (27)). Membrane tortuosity is often estimated using Darcy's law or the KC equations [67]. Direct measurement of τ using 2D cross-sectional images presents a significant challenge as tortuosity is inherently a 3D property. Several relationships have been developed to connect the tortuosity of porous media with other geometric parameters, which leads to estimation of τ for membranes similar to the PVDF_100 as 1.07 to 2.03 in one study [60] and 1.12 to 2.43 in another [68]. Assuming these values, $D_{T,Th}$ for PVDF_100 were calculated using Eq. (27), adopting $D_{f,Th}$ obtained from Method 1. The $D_{T,Th}$ of PVDF_650 was not calculated due to the challenge of finding τ values for similar membranes in literature.

Table 3 also summarizes $D_{T,PSD}$, D_{T,s_BCM} and D_{T,t_BCM} determined using Methods 2–4. The relevant data analysis can be found in Fig. S6 and Fig. S7. The pore size ranges identified for both PSD analysis and the t_BCM are similar to those used for determining D_f, which again confirms the symmetric pore structures of both membranes.

For the same membrane, the trends of D_T values are similar to that of D_f: $D_{T,PSD}$ displays the lowest value, while D_{T,s_BCM} is larger than D_{T,t_BCM}. Interestingly, D_T of PVDF_650 is slightly smaller than that of PVDF_100 obtained using the same method, which is opposite to the trend observed for D_f. This suggests that PVDF_650 has lower τ value than PVDF_100, which is reasonable considering PVDF_650 has significantly larger pore size but similar thickness and porosity. Besides these small discrepancies, the D_T values are comparable between the two membranes, again attributing to the similar pore-formation mechanism involved during the membrane manufacturing.

Lastly, the τ values of both membranes can be calculated using the D_T values obtained from Method 2–4, as listed in Table 3. For PVDF_100, the estimated τ value from $D_{T,PSD}$ is on the larger side of the literature values (again, not from image analysis), while the estimated value from D_{T,t_BCM} is clearly larger.

4.1.3. Permeability of the PVDF membranes

The fractal dimension values (D_f and D_T) obtained above were used to predict the permeability of both membranes (Eq. (23)). We emphasize that the values of λ_{max} and ϕ are both obtained both from SEM image analysis (Table 2), different from other analyses that use pore sizes obtained from porosimetry data. Furthermore, Eq. (23) assumes the $\lambda_{min}/\lambda_{max}$ ratio approaches zero, which is valid for typical fractals. Yu and Li [34] showed that for random porous media with $\lambda_{min}/\lambda_{max} \sim 10^{-2}$, Eq. (23) works well. Here, the $\lambda_{min}/\lambda_{max}$ ratio was 0.091 and 0.094 for PVDF_100 and PVDF_650, respectively (Table 2), indicating a narrower range of self-similarity of the membrane pore structures.

Fig. 5 shows the calculated permeance using the values of D_f and D_T obtained with Methods 1–4 for PVDF_100 and Methods 2–4 for PVDF_650. It is evident that the fractal dimensions obtained from Method 4 (t_BCM) predicted the permeance values that are closest to the experimental data for both membranes. Specifically, the discrepancies between the calculated and experimental values are 6 % for PVDF_100 and 8 % for PVDF_650. For both membranes, fractal dimension values from Model 2 overestimate the permeance values, which is reasonable

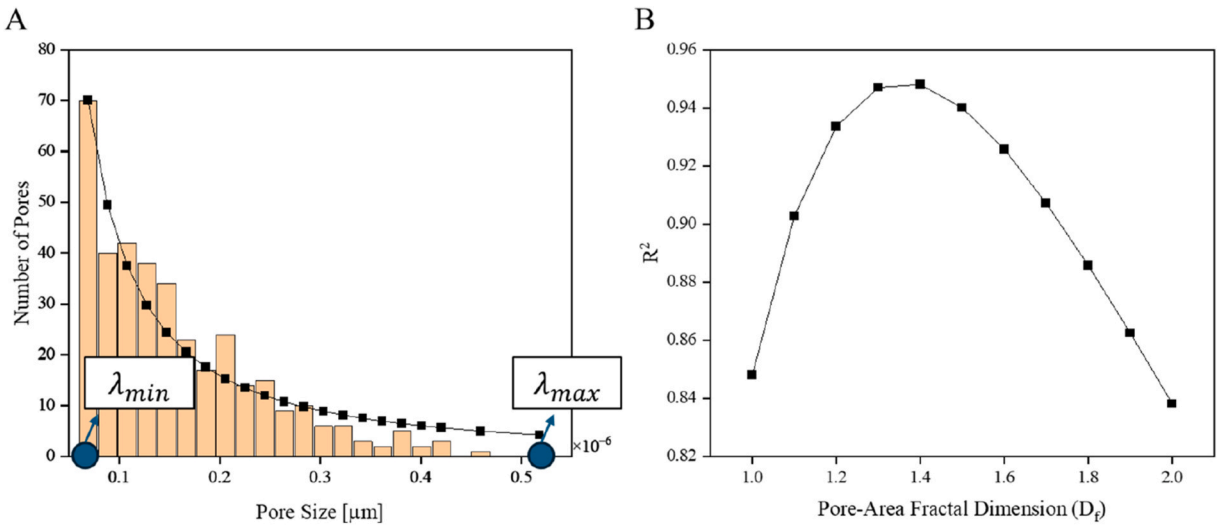


Fig. 3. (A) PSD obtained from the SEM image of PVDF_100. The black curve corresponds to a fit with $R^2 = 0.95$ and a fractal dimension of 1.4. (B) R^2 versus D_f plot.

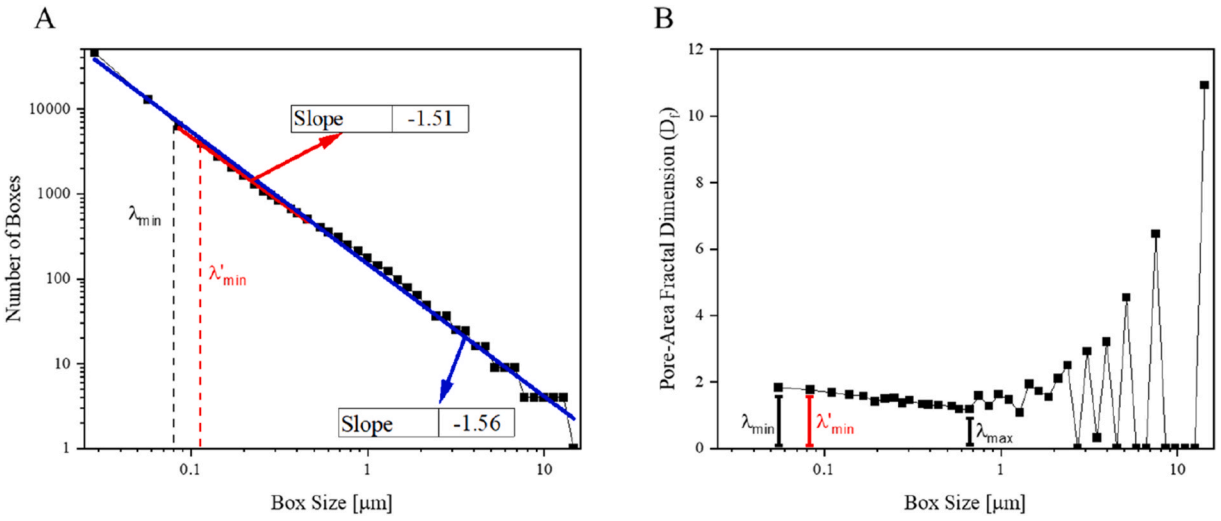


Fig. 4. (A) Number of boxes containing pores versus box size (in μm) obtained from BCM analysis of PVDF_100. The blue line corresponds to the fit across the entire range of box sizes, while the red line corresponds to the fit obtained within the range of minimum and maximum pore sizes. (B) The derivative of the data in (A) at each box size. The range used to extract the D_{f,t_BCM} is marked. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 3
D_T and τ for PVDF_100 and PVDF_650. τ_{Th} values were obtained from literature [60,68].

	$D_{T,Th}$	τ_{Th}	$D_{T,PSD}$	τ_{PSD}	D_{T,s_BCM}	τ_{s_BCM}	D_{T,t_BCM}	τ_{t_BCM}
PVDF_100	1.03 – 1.43	1.07 – 2.43	1.30	2.30	1.75	5.14	1.59	3.61
PVDF_650	–	–	1.10	1.16	1.69	5.10	1.55	2.88

considering this approach provides the lowest D_T (or lowest τ) values. In contrast, fractal dimension values from Method 3 had the largest value of D_T, which leads to underestimation of the membrane permeance.

Based on the above analysis, t_BCM will be used to measure the fractal dimensions of asymmetric PES membranes, as discussed below.

4.2. PES membranes

Asymmetric membranes display a large gradient of pore size (and porosity) throughout the thickness (i.e., throughout the depth of the membrane). The tightest region at the top of the membrane has much higher permeation resistance and dictates the selectivity of the membrane, whereas the majority of the remaining porous membrane has minimal contribution to resistance. Here we performed fractal analysis of two such asymmetric membranes: PES_200 (Fig. 1E and F) and PES_20 (Fig. 1G and H) with nominal pore size ratings of 200 nm and 20 nm, respectively.

For PES_200, λ_{max} increases from 300 to 350 nm at the top surface (Fig. 1E) to >10 μm at the bottom surface, while ϕ increases from 20 % to 80 %. To perform fractal analysis, it is necessary to obtain pore structures at different depths. Accordingly, a laser ablation technique was developed to prepare in-plane cross-sectional images at different depths of PES_200 [41]. Note that the combination of a femtosecond laser with a PP as the pore-preserving agent minimizes the pore

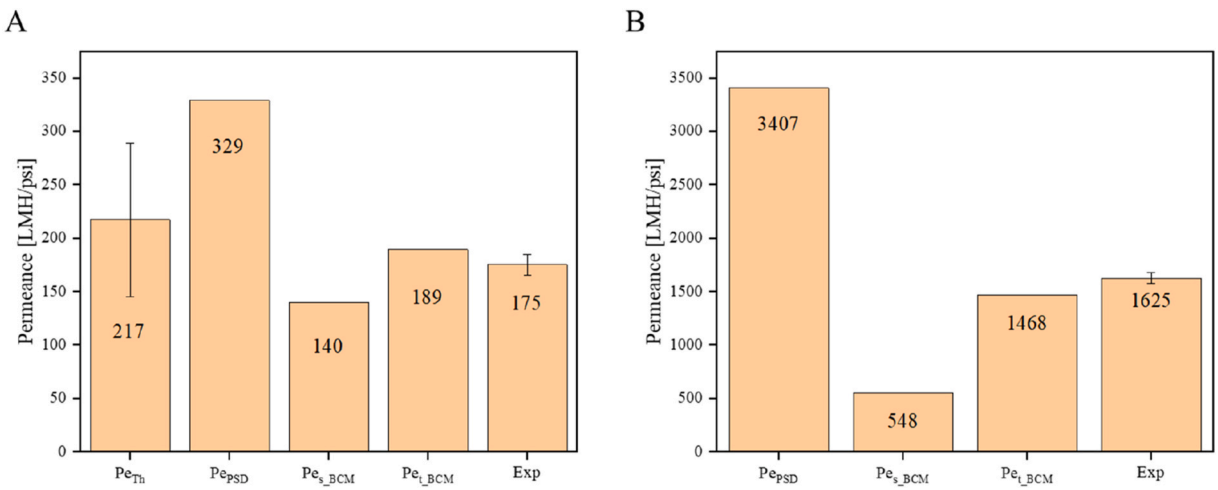


Fig. 5. Water permeance of (A) PVDF_100 and (B) PVDF_650, including permeance values calculated with the different methods of obtaining D_f and D_T, along with experimental results.

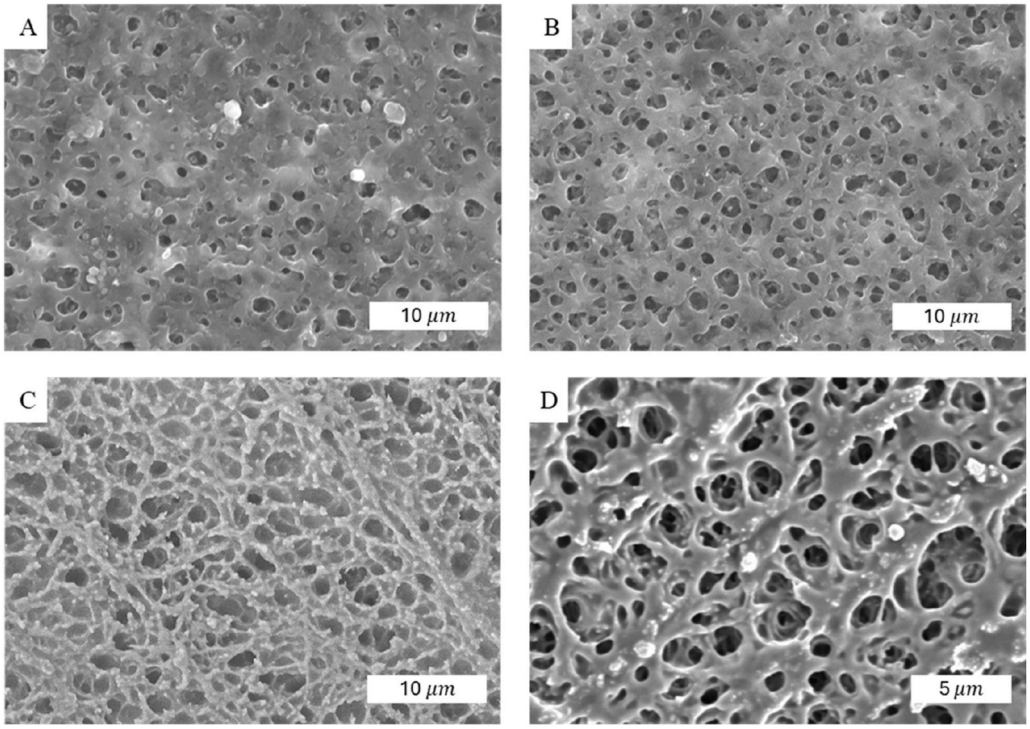


Fig. 6. SEM images of PES_200 at varying ablation depths: (A) 4 μ m, (B) 6 μ m, (C) 40 μ m. (D) 80 μ m.

structure distortion by thermal effect. Fig. 6 shows representative SEM images of PES_200 membrane obtained at 4, 6, 40, and 80 μm depth (measured from the top of the membrane as shown in Fig. 1F).

From the cross-section image of PES_200 (Fig. 1F), we divided the membrane into three layers: a skin layer ($0-10\,\mu m$), an intermediate layer ($10-90\,\mu m$), and a porous sublayer ($90-175\,\mu m$). Furthermore, the skin layer is divided into three layers (each with $3\,\mu m$ thickness) while the intermediate layer is divided into 4 layers (each with $20\,\mu m$ thickness), as shown in Table 4. Note that these layer divisions are somewhat arbitrary. Within each layer, the pore structure is assumed to be symmetric. For example, the SEM images at depths of 4 μm (Fig. 6A) and μm (Fig. 6B) are indeed similar. As such, a single SEM image obtained at the depth within the thickness range of a given layer was used to determine the fractal dimensions of that layer. For example, Fig. 6C and D represent the pore structure of L5 and L7, respectively.

Using this method, the pore structure parameters of each layer (λ_{max}, ϕ, D_f and D_T) are determined, as summarized in Table 4. Note that for the fractal analysis of each layer, Eq. (12) needs to be satisfied. For the eight layers, $\left(\frac{\lambda_{min}}{\lambda_{max}}\right)^{D_f}$ value decreases from 0.054 (L1) to 0.0024 (L8). These values are certainly not infinitely small, but their contribution to Eq. (17) is negligible. As expected, both λ_{max} and ϕ increase with increase of layer number (or depth). Furthermore, both D_f and D_T values also increase with the increase of depth. In particular, D_f increases from 1.27 to 1.88, which correlates with the increase of ϕ (Eq. (25)). We emphasize that the increase in D_T does not indicate an increase of membrane tortuosity (τ). On the contrary, as calculated using Eq. (15), τ decreases from ~4.0 for the skin layer to ~1.0 for the porous sublayer (Fig. 7A). The proportionality between D_T and τ is only valid when λ_{max}

Table 4
Pore structure parameters for the 8 layers of PES_200 membrane.

PES_200	l_0 [μ m]	D_f	D_T	ϕ [%]	λ_{max} [μm]	$Pe \times 10^3$ [LMH/psi]	C_i [%]
Top Skin Layer							
L1	0 – 3	1.27	1.3	20	0.33	1.29	66.1
L2	4 – 6	1.45	1.41	26	0.70	5.80	14.5
L3	7–9	1.57	1.66	35	0.83	7.67	10.0
Intermediate layer							
L4	10 – 29	1.61	1.67	41	2.92	17.89	4.7
L5	30 – 49	1.73	1.74	53	3.37	25.36	3.0
L6	50 – 69	1.77	1.75	66	4.35	73.48	1.1
L7	70–89	1.81	1.81	72	5.30	119.40	0.6
Porous Sublayer							
L8	>90	1.88	1.86	80	15.00	1090.15	0.1

and ϕ remain constant.

Using the pore structure parameters the permeability of each layer was calculated using Eq. (23). Next, permeance of each layer (Pe) was calculated, as listed in Table 4. The overall permeance of the PES_200 (Pe_{tot}) was then calculated using a resistance-in-series model,

$$Pe_{tot} = \frac{1}{R_{tot}} = \frac{1}{\sum_{i=1}^{8} R_i} = \frac{1}{\sum_{i=1}^{8} \frac{1}{Pe_i}} \tag{28}$$

where R_i (and Pe_i) is the resistance (and permeance) of L1 to L8, and R_{tot} is the overall resistance of the membrane. The Pe_{tot} obtained for PES_200 is 836 LMH/psi, which matches reasonably well with the experimental value of 920 ± 72 LMH/psi.

Fig. 7B plots the resistance of the eight layers throughout the membrane thickness. As expected, the top skin layer (L1-L3) has the largest contribution to the overall resistance of the membrane. The specific contribution of each layer to the overall resistance is estimated,

$$C_i = \left(\frac{R_i}{R_{tot}}\right) \times 100 \tag{29}$$

As listed in Table 4 and displayed in Fig. 7B, the first three layers (top 10 μm) contribute more than 90 % of the total resistance, the intermediate porous sublayer contributes less than 10 %, whereas the porous sublayer contributes less than 1 %. Note that the above analysis can be certainly improved if the full 3D images of the asymmetric porous structure were available to allow better definition of the layers. However, we do not expect it will significantly change the resistance-thickness trend shown in Fig. 7B.

4.3. PES_20 membrane

A similar analysis was carried out for PES_20. Here, the depth-dependent images were obtained using FIB-SEM. Based on the analysis above, only the top 2.5 μ m layer of PES_20 was examined, which is the region that most of the membrane resistance originates from. Fig. 8 shows representative 2D SEM images of the PES_20 at varying depths, sliced from the 3D FIB-SEM images. The porosity of the membrane increases with thickness from ~8 % at the surface to ~34 % after reaching a thickness above 0.5 μm. The overall (average) porosity for the first 2 μ m layer is 32.9 %.

The 2-μm layer membrane was then divided into 4 layers, each representing successive 0.5-μm layers (Fig. 9A). Using t_BCM, D_f and D_T values for each layer were obtained, while λ_{max} and ϕ are determined directly from the images. Note that D_T values obtained from 3 locations in each layer (Fig. 9A) showed less than 5 % variation. As listed in Table 5, λ_{max}, ϕ, and D_f increase with the increase of depth. In comparison, D_T decreases from 1.80 (L1), to 1.75 (L2), which corresponds to a decrease of membrane tortuosity from 6.9 to 2.1 (Eq. (15)), all of which are larger than that of PES_200. Note that typical τ values reported for membranes are in the range of 1–3 for the entire membrane [10,69]. Here, the higher tortuosity (>3) corresponds to the top 1 μm skin of the membrane, which shows a more complex interconnectivity.

Using Eq. (23), the permeability, and subsequently, permeance of each layer was calculated (Table 5). The overall permeance of the top 2 μm layer of PES_20, calculated based on Eq. (28), was 59 LMH/psi, compared to 40 LMH/psi [44]. Certainly, the remaining membrane thickness will have some additional resistance, which will reduce the estimated permeance. Nevertheless, the analysis shows that the first 2 μ m of the PES_20 contributes to the majority of the overall membrane resistance (>67 %). Clearly, PES_20 has a denser and thinner skin layer than PES_200.

5. Conclusion

In this study, we present systematic quantifications of membrane pore structures using fractal analysis. Using two symmetric PVDF membranes, four different methods of extracting fractal dimensions from SEM images, including both pore-area (D_f) and tortuosity fractal

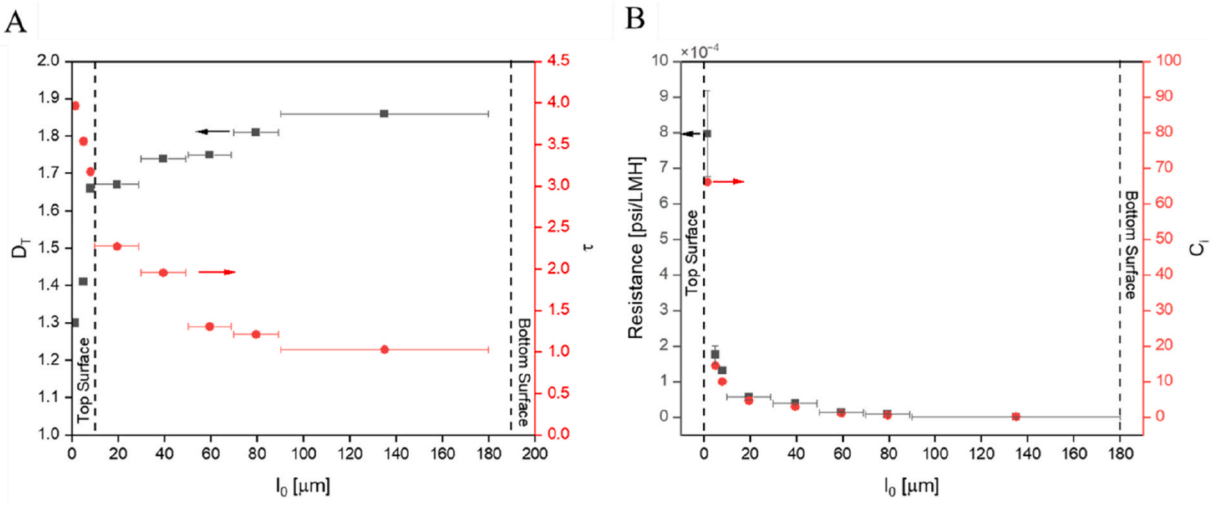


Fig. 7. (A) D_T (determined using t_BCM) and τ (calculated from Eq. (15)) for layers L1 to L8. (B) Permeation resistance (left axis) and corresponding percentage to total membrane resistance (right axis) of different layers throughout the entire membrane thickness. Note that error bars for x-axis represent the range of thickness for the layers.

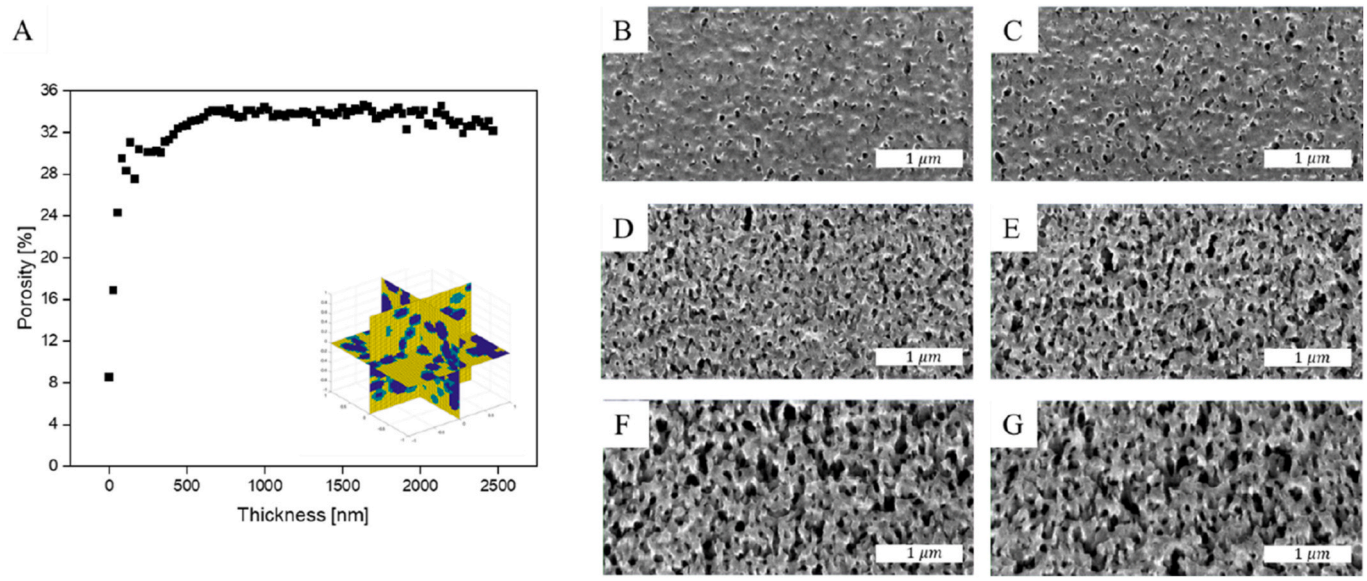


Fig. 8. (A) 2D Porosity of PES_20 at different depths, obtained from the in-plane images. Representative SEM images at (B) surface, (C) 14 nm depth, (D) 570 nm depth, (E) 1 μm depth, (F) 2 μm depth, and (G) 2.5 μ m depth.

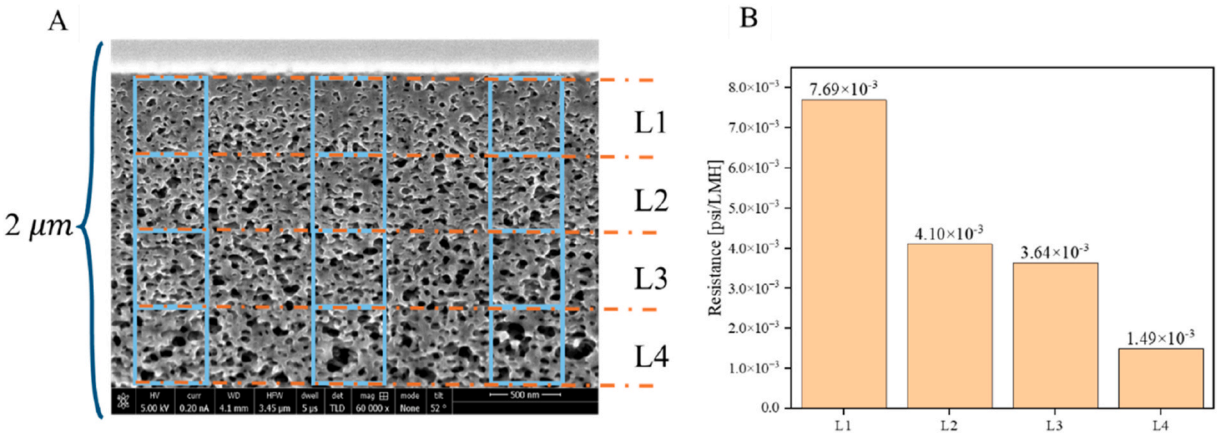


Fig. 9. (A) Cross-sectional images of PES_20 showing the 4 layers (L1-L4). In each layer, fractal dimensions of three regions (boxed) were measured. (B) Calculated resistance of layers L1-L4.

Table 5
Pore structure parameters and permeance for the four layers of PES_20.

	Thickness [μm]	λ_{max} [nm]	$\phi_{average}$ [%]	D_f	D_T	τ	Pe [LMH/psi]
L1	0.50	45	28.5	1.41	1.80	6.88	130
L2	0.50	72	31.4	1.56	1.81	3.86	244
L3	0.50	89	31.9	1.68	1.82	2.21	275
L4	0.50	110	33.0	1.63	1.75	2.11	672

dimensions (D_T), are compared. The study reveals that BCM, which is commonly used to determine fractal dimensions of porous media, provides better estimations using a box size range between the minimum and maximum pore size of the membranes. Most significantly, the permeability of the membranes can be calculated based only on geometric parameters (λ_{max}, ϕ, D_f, and D_T), all of which can be obtained from 2D membrane images. For both PVDF membranes, the calculated values match well with the experimental results.

The method was further applied to two asymmetric PES membranes that have depth-dependent pore structures. SEM images of PES_200 and PES_20 at different depths were obtained using a controlled laser ablation method and FIB-SEM, respectively. Pore geometric parameters (λ_{max}, ϕ, D_f, and D_T) were determined at multiple depths, which allows estimation of permeability of the porous layers. The analysis shows that the tight top region of the membrane, 3 μm for PES_200, and 2 μm for PES_20, contribute more than 66 % of the overall membrane resistance. Using the resistance-in-series approach, the calculated overall membrane permeance matches well with the experimental results for both PES membranes.

The study demonstrated the potential of fractal analysis to quantify pore structures of membranes using images. Certainly, the method involves uncertainties from sample preparation (laser ablation and FIB-SEM) and imaging processing, which requires additional work for error quantifications. On the other hand, improvements in these procedures (e.g. reliable images of high-resolution 3D pore structures) will help to improve the accuracy of the analysis.

CRediT authorship contribution statement

Riccardo Lovison: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Mohammad A. Afzal:** Writing – review & editing, Methodology, Data

curation. **Christina M. Carbrello:** Writing – review & editing, Resources, Project administration, Investigation, Data curation. **Andrew L. Zydney:** Writing – review & editing, Data curation. **Yifu Ding:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We acknowledge the support from the National Science Foundation (NSF) Industry/University Cooperative Research Center for Membrane Application, Science and Technology (MAST) at the University of Colorado Boulder (UCB, award number, EEC 2310937).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124168.

Data availability

Data will be made available on request.

References

[1] R. van Reis, A. Zydney, Membrane separations in biotechnology, Curr. Opin. Biotechnol. 12 (2) (2001) 208–211, https://doi.org/10.1016/S0958-1669(00)00201-9.
[2] N. Taylor, W.J. Ma, A. Kristopeit, S. Wang, A.L. Zydney, Retention characteristics of Sterile filters – effect of pore size and structure, J. Membr. Sci. 635 (2021) 119436, https://doi.org/10.1016/j.memsci.2021.119436.
[3] H. Chamani, J. Woloszyn, T. Matsuura, D. Rana, C.Q. Lan, Pore wetting in membrane distillation: a comprehensive review, Prog. Mater. Sci. 122 (2021) 100843, https://doi.org/10.1016/j.pmatsci.2021.100843.
[4] J. Phattaranawik, R. Jiraratananon, A.G. Fane, Effect of pore size distribution and air flux on mass transport in direct contact membrane distillation, J. Membr. Sci. 215 (1) (2003) 75–85, https://doi.org/10.1016/S0376-7388(02)00603-8.
[5] A.O. Imdakm, T. Matsuura, Simulation of heat and mass transfer in direct contact membrane distillation (MD): the effect of membrane physical properties, J. Membr. Sci. 262 (1) (2005) 117–128, https://doi.org/10.1016/j.memsci.2005.05.026.
[6] H. Chamani, A. Rabbani, K.P. Russell, A.L. Zydney, E.D. Gomez, J. Hattrick-Simpers, J.R. Werber, Data-science-based reconstruction of 3-D membrane pore structure using a single 2-D micrograph, J. Membr. Sci. 678 (2023) 121673, https://doi.org/10.1016/j.memsci.2023.121673.
[7] A. Ley, P. Altschuh, V. Thom, M. Selzer, B. Nestler, P. Vana, Characterization of a macro porous polymer membrane at micron-scale by confocal-laser-scanning microscopy and 3D image analysis, J. Membr. Sci. 564 (2018) 543–551, https://doi.org/10.1016/j.memsci.2018.07.062.
[8] M.M. Kłosowski, C.M. McGilvery, Y. Li, P. Abellan, Q. Ramasse, J.T. Cabral, A. G. Livingston, A.E. Porter, Micro-to nano-scale characterisation of polyamide structures of the SW30HR RO membrane using advanced electron microscopy and stain tracers, J. Membr. Sci. 520 (2016) 465–476, https://doi.org/10.1016/j.memsci.2016.07.063.
[9] A. Hernández, J.I. Calvo, P. Prádanos, F. Tejerina, Pore size distributions in microporous membranes. A critical analysis of the bubble point extended method, J. Membr. Sci. 112 (1) (1996) 1–12, https://doi.org/10.1016/0376-7388(95)00025-9.
[10] Md A. Islam, M. Ulbricht, Microfiltration membrane characterization by gas-liquid displacement porometry: matching experimental pore number distribution with liquid permeability and bulk porosity, J. Membr. Sci. 569 (2019) 104–116, https://doi.org/10.1016/j.memsci.2018.09.030.
[11] S. Giglia, D. Bohonak, P. Greenhalgh, A. Leahy, Measurement of pore size distribution and prediction of membrane filter virus retention using liquid–liquid porometry, J. Membr. Sci. 476 (2015) 399–409, https://doi.org/10.1016/j.memsci.2014.11.053.
[12] E. Honold, E.L. Skau, Application of mercury-intrusion method for determination of pore-size distribution to membrane filters, Science 120 (3124) (1954) 805–806, https://doi.org/10.1126/science.120.3124.805.
[13] G. Reichelt, Bubble point measurements on large areas of microporous membranes, J. Membr. Sci. 60 (2) (1991) 253–259, https://doi.org/10.1016/S0376-7388(00)81538-0.
[14] J.I. Calvo, A. Bottino, G. Capannelli, A. Hernández, Comparison of liquid–liquid displacement porosimetry and scanning electron microscopy image analysis to characterise ultrafiltration track-etched membranes, J. Membr. Sci. 239 (2) (2004) 189–197, https://doi.org/10.1016/j.memsci.2004.02.038.
[15] P. Kosiol, B. Hansmann, M. Ulbricht, V. Thom, Determination of pore size distributions of virus filtration membranes using gold nanoparticles and their correlation with virus retention, J. Membr. Sci. 533 (2017) 289–301, https://doi.org/10.1016/j.memsci.2017.03.043.
[16] R.I. Peinador, J.I. Calvo, K. ToVinh, V. Thom, P. Prádanos, A. Hernández, Liquid–liquid displacement porosimetry for the characterization of virus retentive membranes, J. Membr. Sci. 372 (1) (2011) 366–372, https://doi.org/10.1016/j.memsci.2011.02.022.
[17] B. Sutariya, S. Karan, A realistic approach for determining the pore size distribution of nanofiltration membranes, Sep. Purif. Technol. 293 (2022) 121096, https://doi.org/10.1016/j.seppur.2022.121096.
[18] A.L. Zydney, P. Aimar, M. Meireles, J.M. Pimbley, G. Belfort, Use of the log-normal probability density function to analyze membrane pore size distributions: functional forms and discrepancies, J. Membr. Sci. 91 (3) (1994) 293–298, https://doi.org/10.1016/0376-7388(94)80090-1.
[19] A. Mehta, A.L. Zydney, Permeability and selectivity analysis for ultrafiltration membranes, J. Membr. Sci. 249 (1) (2005) 245–249, https://doi.org/10.1016/j.memsci.2004.09.040.
[20] M.U. Siddiqui, A.F.M. Arif, S. Bashmal, Permeability-selectivity analysis of microfiltration and ultrafiltration membranes: effect of pore size and shape distribution and membrane stretching, Membranes 6 (3) (2016) 40, https://doi.org/10.3390/membranes6030040.
[21] M. Rehman, M.B. Hafeez, M. Krawczuk, A comprehensive review: applications of the kozeny–carman model in engineering with permeability dynamics, Arch. Comput. Methods Eng. (2024), https://doi.org/10.1007/s11831-024-10094-7.
[22] B.B. Mandelbrot, A. Blumen, M. Fleischmann, D.J. Tildesley, R.C. Ball, Fractal geometry: what is It, and what does it do? Proc. R. Soc. Lond. Math. Phys. Sci. 423 (1864) (1997) 3–16, https://doi.org/10.1098/rspa.1989.0038.
[23] J. Wu, B. Yu, A fractal resistance model for flow through porous media, Int. J. Heat Mass Tran. 50 (19) (2007) 3925–3932, https://doi.org/10.1016/j.ijheatmasstransfer.2007.02.009.
[24] P. Xu, B. Yu, Developing a new form of permeability and kozeny–carman constant for homogeneous porous media by means of fractal geometry, Adv. Water Resour. 31 (1) (2008) 74–81, https://doi.org/10.1016/j.advwatres.2007.06.003.
[25] J. Cai, B. Yu, Prediction of maximum pore size of porous media based on fractal geometry, Fractals 18 (4) (2010) 417–423, https://doi.org/10.1142/S0218348X10005123.
[26] J. Cai, B. Yu, M. Zou, L. Luo, Fractal characterization of spontaneous Co-Current imbibition in porous media, Energy Fuels 24 (3) (2010) 1860–1867, https://doi.org/10.1021/ef901413p.
[27] G. He, Z. Zhao, P. Ming, A. Abuliti, C. Yin, A fractal model for predicting permeability and liquid water relative permeability in the gas diffusion layer (GDL) of PEMFCs, J. Power Sources 163 (2) (2007) 846–852, https://doi.org/10.1016/j.jpowsour.2006.09.054.
[28] L.-Z. Zhang, A fractal model for gas permeation through porous membranes, Int. J. Heat Mass Tran. 51 (21) (2008) 5288–5295, https://doi.org/10.1016/j.ijheatmasstransfer.2008.03.008.
[29] M.S. Win, K.I. Winey, A.L. Frischknecht, Morphology–diffusivity relationships in fluorine-free random terpolymers for proton-exchange membranes, Macromolecules 56 (23) (2023) 9905–9913, https://doi.org/10.1021/acs.macromol.3c01707.
[30] J. Martinez, M. Aghajani, Y. Lu, A.K. Blevins, S. Fan, M. Wang, J.P. Killgore, S. B. Perez, J. Patel, C. Carbrello, S. Foley, R. Sylvia, R. Long, R. Castro, Y. Ding, Capillary bonding of membranes by viscous polymers: infiltration kinetics and mechanical integrity of the bonded polymer/membrane structures, J. Membr. Sci. 641 (2022) 119898, https://doi.org/10.1016/j.memsci.2021.119898.
[31] B.B. Mandelbrot, The Fractal Geometry of Nature, Henry Holt and Company, 1983.
[32] G.A. Losa, D. Ristanović, D. Ristanović, I. Zaletel, S. Beltraminelli, From fractal geometry to fractal analysis, Appl. Math. 7 (4) (2016) 346–354, https://doi.org/10.4236/am.2016.74032.
[33] J. Feder, Fractals, Springer Science & Business Media, 2013.
[34] B. Yu, J. Li, Some fractal characters of porous media, Fractals 9 (3) (2001) 365–372, https://doi.org/10.1142/S0218348X01000804.
[35] A. Majumdar, B. Bhushan, Role of fractal geometry in roughness characterization and contact mechanics of surfaces, J. Tribol. 112 (2) (1990) 205–216, https://doi.org/10.1115/1.2920243.
[36] S.W. Wheatcraft, S.W. Tyler, An explanation of scale-dependent dispersivity in heterogeneous aquifers using concepts of fractal geometry, Water Resour. Res. 24 (4) (1988) 566–578, https://doi.org/10.1029/WR024i004p00566.
[37] Y. Shi, J. Xiao, M. Pan, R. Yuan, A fractal permeability model for the gas diffusion layer of PEM fuel cells, J. Power Sources 160 (1) (2006) 277–283, https://doi.org/10.1016/j.jpowsour.2006.01.032.
[38] X. Shen, L. Li, W. Cui, Y. Feng, Improvement of fractal model for porosity and permeability in porous materials, Int. J. Heat Mass Tran. 121 (2018) 1307–1315, https://doi.org/10.1016/j.ijheatmasstransfer.2018.01.084.
[39] B. Yu, W. Liu, Fractal analysis of permeabilities for porous media, AIChE J. 50 (1) (2004) 46–57, https://doi.org/10.1002/aic.10004.
[40] H. Kaya, S. Fan, S.H. Kim, Y. Ding, B.D. Vogt, IR spectroscopic ellipsometry to characterize microfiltration membranes, ACS Appl. Polym. Mater. 5 (6) (2023) 3928–3937, https://doi.org/10.1021/acsapm.3c00083.
[41] M. Aghajani, S.H. Maruf, M. Wang, J. Yoshimura, G. Pichorim, A. Greenberg, Y. Ding, Relationship between permeation and deformation for porous membranes,

J. Membr. Sci. 526 (2017) 293–300, https://doi.org/10.1016/j.memsci.2016.12.048.
[42] J. Martinez, S. Fan, S. Rabade, A.K. Blevins, K. Fung, J.P. Killgore, S.B. Perez, K. Youngbear, C. Carbrello, S. Foley, X. Ding, R. Long, R. Castro, Y. Ding, Capillary infiltration kinetics in highly asymmetric porous membranes and the resulting debonding behaviors, Polymer 263 (2022) 125529, https://doi.org/10.1016/j.polymer.2022.125529.
[43] H.H. Wang, J.T. Jung, J.F. Kim, S. Kim, E. Drioli, Y.M. Lee, A novel green solvent alternative for polymeric membrane preparation via nonsolvent-induced phase separation (NIPS), J. Membr. Sci. 574 (2019) 44–54, https://doi.org/10.1016/j.memsci.2018.12.051.
[44] K.P. Brickey, A.L. Zydney, E.D. Gomez, FIB-SEM tomography reveals the nanoscale 3D morphology of virus removal filters, J. Membr. Sci. 640 (2021) 119766, https://doi.org/10.1016/j.memsci.2021.119766.
[45] Lovison, R.; Kabutz, H.; Letourneau, E.; Fung, K.; Jain, R.; Carbrello, C.; Kaushik, J.; Ding, Y. Femtosecond Laser Ablation of Porous PVDF and PES Membranes: Ablation Behaviors and In-Plane Cross-Sectioning. **Manuscript under review**.
[46] C.A. Schneider, W.S. Rasband, K.W. Eliceiri, NIH image to ImageJ: 25 years of image analysis, Nat. Methods 9 (7) (2012) 671–675, https://doi.org/10.1038/nmeth.2089.
[47] Q. Tseng, I. Wang, E. Duchemin-Pelletier, A. Azioune, N. Carpi, J. Gao, O. Filhol, M. Piel, M. Théry, M. Balland, A new micropatterning method of soft substrates reveals that different tumorigenic signals can promote or reduce cell contraction levels, Lab Chip 11 (13) (2011) 2231–2240, https://doi.org/10.1039/C0LC00641F.
[48] D. Bradley, G. Roth, Adaptive thresholding using the integral image, J. Graph. Tool. 12 (2) (2007) 13–21, https://doi.org/10.1080/2151237X.2007.10129236.
[49] J. Li, Q. Du, C. Sun, An improved box-counting method for image fractal dimension estimation, Pattern Recogn. 42 (11) (2009) 2460–2469, https://doi.org/10.1016/j.patcog.2009.03.001.
[50] K. Foroutan-pour, P. Dutilleul, D.L. Smith, Advances in the implementation of the box-counting method of fractal dimension estimation, Appl. Math. Comput. 105 (2) (1999) 195–210, https://doi.org/10.1016/S0096-3003(98)10096-6.
[51] C. Panigrahy, A. Seal, N.K. Mahato, D. Bhattacharjee, Differential box counting methods for estimating fractal dimension of gray-scale images: a survey, Chaos Solitons Fractals 126 (2019) 178–202, https://doi.org/10.1016/j.chaos.2019.06.007.
[52] Y. Liu, L. Chen, H. Wang, L. Jiang, Y. Zhang, J. Zhao, D. Wang, Y. Zhao, Y. Song, An improved differential box-counting method to estimate fractal dimensions of gray-level images, J. Vis. Commun. Image Represent. 25 (5) (2014) 1102–1111, https://doi.org/10.1016/j.jvcir.2014.03.008.
[53] N. Sarkar, B.B. Chaudhuri, An efficient differential box-counting approach to compute fractal dimension of image, IEEE Trans. Syst. Man Cybern. 24 (1) (1994) 115–120, https://doi.org/10.1109/21.259692.
[54] J.E. Hutchinson, Fractals and self similarity, Ind. Univ. Math. J. 30 (5) (1981) 713–747.
[55] D.W. Boyd, The residual set dimension of the apollonian packing, Mathematika 20 (2) (1973) 170–174, https://doi.org/10.1112/S0025579300004745.
[56] M.A. Reiss, N. Sabathiel, H. Ahammer, Noise dependency of algorithms for calculating fractal dimensions in digital images, Chaos Solitons Fractals 78 (2015) 39–46, https://doi.org/10.1016/j.chaos.2015.07.004.
[57] W. Jiang, Y. Liu, J. Wang, R. Li, X. Liu, J. Zhang, Problems of the grid size selection in differential box-counting (DBC) methods and an improvement strategy, Entropy 24 (7) (2022) 977, https://doi.org/10.3390/e24070977.
[58] A. Dathe, M. Thullner, The relationship between fractal properties of solid matrix and pore space in porous media, Geoderma 129 (3) (2005) 279–290, https://doi.org/10.1016/j.geoderma.2005.01.003.
[59] E. Quartarone, P. Mustarelli, A. Magistris, Transport properties of porous PVDF membranes, J. Phys. Chem. B 106 (42) (2002) 10828–10833, https://doi.org/10.1021/jp0139843.
[60] I.V. Curcino, P.R.S.C. Júnior, A.O.C. Gómez, L.E.P. Chenche, J.A. Lima, C. P. Naveira-Cotta, R.M. Cotta, Analysis of effective thermal conductivity and tortuosity modeling in membrane distillation simulation, Micro Nano. Eng. 17 (2022) 100165, https://doi.org/10.1016/j.mne.2022.100165.
[61] A. Adeniyi, R. Mbaya, P. Popoola, F. Gomotsegang, I. Ibrahim, M. Onyango, Predicting the fouling tendency of thin film composite membranes using fractal analysis and membrane autopsy, Alex. Eng. J. 59 (6) (2020) 4397–4407, https://doi.org/10.1016/j.aej.2020.07.046.
[62] F. Meng, H. Zhang, Y. Li, X. Zhang, F. Yang, Application of fractal permeation model to investigate membrane fouling in membrane bioreactor, J. Membr. Sci. 262 (1) (2005) 107–116, https://doi.org/10.1016/j.memsci.2005.04.013.
[63] L. Cipelletti, M. Carpineti, M. Giglio, Fractal morphology, spatial order, and pore structure in microporous membrane filters, Langmuir 12 (26) (1996) 6446–6451, https://doi.org/10.1021/la9607363.
[64] M. Song, B. Jiang, Dynamic scaling and fractal behaviour of spinodal phase separation in polymer mixtures of Poly(Methyl methacrylate) with Poly(Styrene-Co-Acrylonitrile), Polymer 33 (7) (1992) 1445–1448, https://doi.org/10.1016/0032-3861(92)90120-L.
[65] X. Li, W. Wei, L. Wang, J. Cai, Fractal dimension of digital 3D rock models with different pore structures, Energies 15 (20) (2022) 7461, https://doi.org/10.3390/en15207461.
[66] Q. Wu, B. Wu, Study of membrane morphology by image analysis of electron micrographs, J. Membr. Sci. 105 (1) (1995) 113–120, https://doi.org/10.1016/0376-7388(95)00055-H.
[67] P.C. Carman, Fluid flow through granular beds, Chem. Eng. Res. Des. 75 (1997) S32–S48, https://doi.org/10.1016/S0263-8762(97)80003-2.
[68] W.-J. Kim, O. Campanella, D.R. Heldman, Predicting the performance of direct contact membrane distillation (DCMD): mathematical determination of appropriate tortuosity based on porosity, J. Food Eng. 294 (2021) 110400, https://doi.org/10.1016/j.jfoodeng.2020.110400.
[69] S.-H. Lee, W.-S. Chang, S.-M. Han, D.-H. Kim, J.-K. Kim, Synchrotron X-Ray nanotomography and three-dimensional nanoscale imaging analysis of pore structure-function in nanoporous polymeric membranes, J. Membr. Sci. 535 (2017) 28–34, https://doi.org/10.1016/j.memsci.2017.04.024.